European Journal of Sport Science

ISSN: 1746-1391 (Print) 1536-7290 (Online) Journal homepage: https://www.tandfonline.com/loi/tejs20

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To cite this article: Erin H. Feser, Paul Macadam & John B. Cronin (2020) The effects of lower limb wearable resistance on sprint running performance: A systematic review, European Journal of Sport Science, 20:3, 394-406, DOI: 10.1080/17461391.2019.1629631

To link to this article: https://doi.org/10.1080/17461391.2019.1629631

Accepted author version posted online: 10 Jun 2019.
Published online: 02 Jul 2019.

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European Journal of Sport Science, 2020
Vol. 20, No. 3, 394–406, https://doi.org/10.1080/17461391.2019.1629631

The effects of lower limb wearable resistance on sprint running performance: A systematic review

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ABSTRACT
The aim of this review was to examine the literature that has used lower limb wearable resistance (WR) during sprint running. A systematic search was completed to identify acute and longitudinal studies assessing the effects of lower limb WR on sprint running performance from international peer-reviewed journals. The Boolean phrases (limb OR leg OR lower extremity) AND (sprint*) AND (resist* OR weight OR load*) were used to search PubMed, SPORTDiscus, and Web of Science electronic databases. Ten studies met the inclusion criteria and were retained for analysis that reported the acute kinematic and kinetic effects ($n = 8$), acute performance effects ($n = 3$), and longitudinal effects ($n = 1$). Results showed that the WR micro-loading (0.6–5% body mass) significantly increased contact time (2.9–8.9%), decreased step frequency (−1.4 to −3.7%), and slowed total sprint times (0.6–7.4%). Unloaded sprinting immediately following sprints with lower limb WR resulted in no significant change to total sprinting times. One longitudinal training study did not find a significant effect on maximal sprinting speed for non-trained participants. It can be concluded that not all step kinematic variables are affected during sprinting with an added load up to 5% body mass. Therefore, coaches can use lower limb WR to selectively overload certain aspects of sprint running, in particular stride frequency. It also appears that lower limb WR overloads sprint movement velocity and may provide a stimulus to increase horizontal force output, therefore, it may be inferred that lower limb WR has the potential to elicit improved sprinting performance.

Keywords: *Kinematics, kinetics, acceleration, maximal velocity, resistance, specificity*

Highlights
- Lower limb wearable resistance training involves attaching an external micro-load to the lower limbs and wearing the load during sport specific movement training.
- Although a concern with this type of training is that the added load can unfavorably change the sprint running movement pattern, study results suggest that stride length is unchanged when sprint running with lower limb wearable resistance.
- This form of resistance training may be used to target certain aspects of sprint running such as stride frequency and provide a stimulus to increase horizontal force output for short distance sprint running or for the acceleration phase of longer distance sprint running.

Introduction

Sprint running ability is a key performance factor in many sport activities such as track and field, rugby, and football. Several training options are available to produce speed adaptation, however this adaptation needs to be specific to the sport and athlete requirements (Baker, 1996). Though non-specific training plays a role in certain phases of a periodised plan, the transference of non-specific strength and power to speed and agility is usually minimal at best (Cronin, Ogden, Lawton, & Brughelli, 2007). Based on the principle of training specificity, training options should replicate the characteristics of a sporting action so that training adaptations will optimally transfer to the sporting action (Cronin & Hansen, 2006). One training method that enables such specificity for improving sprint performance is wearable resistance (WR) training. WR training involves attaching an external micro-load (i.e. loads as little as 0.5% body mass (BM)) to different segments of the body. The load is worn during sport specific movement training and is, therefore, a direct example of the

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application of the concept of training specificity (i.e. resisted movement training) (Macadam, Cronin, & Simperingham, 2017a). Previous sprint WR studies have used loads attached to the upper-limbs (Macadam, Simperingham, & Cronin, 2018; McNaughton & Kelly, 2010; Ropret, Kukolj, Ugarkovic, Matavulj, & Jaric, 1998), trunk (Clark, Stearne, Walts, & Miller, 2010; Cronin, Hansen, Kawamori, & Mcnair, 2008; Cross, Brughelli, & Cronin, 2014), or lower limbs (Macadam, Simperingham, & Cronin, 2017b; Simperingham & Cronin, 2014). These forms of resisted sprint training attempt to match the sprint training programme design goals described by Cissik (2004) of increasing neural activation and strength of the lower limbs, thus sprint velocity, without adversely affecting sprint technique.

The ability to load specific joints, and therefore specific muscles, by attaching external micro-loads to an athlete's limb(s) makes WR training, and in particular lower limb WR, an attractive option for sprint training. However, attaching an external load directly onto the limb changes the limb's inertial properties. These changes are more prominent when the load is positioned in a more distal location on the limb and when load magnitude increases (Martin & Cavanagh, 1990). As the inertial properties of the limb are changed, there are accompanying biomechanical changes (Martin & Cavanagh, 1990). Practitioners interested in utilising lower limb WR training to develop an athlete's sprint running capabilities must first understand the biomechanical changes that occur with limb loading while sprinting. Though WR sprint training has increased in popularity, there is currently limited evidence supporting the choice of load placement and magnitude for training application. The aim of this review was to evaluate the literature that has used lower limb WR during sprint running to provide the practitioner with a summary and critique of the current scientific knowledge in this area.

Methods

A systematic search, completed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses statement guidelines (Moher, Liberati, Tetzlaff, & Altman, 2009), was completed to identify research that had quantified the acute and longitudinal effects of lower limb WR on sprint running performance. The Boolean phrases (limb OR leg OR lower extremity) AND (sprint*) AND (resist* OR weight OR load*) were used for searches in PubMed, SPORTDiscus, and Web of Science electronic databases from inception to November 2018. Following, any additional studies that met the inclusion criteria previously known by the authors or identified from the reference lists of the retrieved studies were also included.

Studies measuring the kinetic or kinematic effects of lower limb WR during sprint running that were published in peer reviewed journals were included. The lower limb WR included loads attached to the thigh, shank, or foot, or any combination thereof. Additionally, studies that assessed sprint running immediately following WR use during sprint running (revealing acute performance effects) or following a WR training intervention were also included. Studies were excluded when the loads were not directly attached to the body, when sprint running was not performed at maximum effort (e.g. running speeds), and when results could not be extracted as numerical data (e.g. figures). Further, only the studies that included injury free participants, regardless of age, gender, or training status were included. If the same study was published in multiple locations, one was retained while the others were considered a duplication and removed (e.g. Pajic, 2011; Pajic, Kostovski, Ilic, Jakovljevic, & Preljevic, 2011).

The outcome of search results from the electronic databases, reference list reviews, and prior knowledge resulted in 692 relevant studies (Figure 1). Following application of the inclusion and exclusion criteria, ten studies were included for review.

Results

Ten studies were analysed in this review, which consisted of 116 participants that completed a sprint running intervention with lower limb WR. A wide range of training status and sporting experience was included, ranging from healthy, untrained individuals to semi-professional and national level athletes. Cohen's *d* effect sizes (ES) were calculated for the variables reported as significant by the studies included in this review and were described as small (<0.5), moderate (0.51–0.79) and large (>0.8) (Cohen, 1988). The following sections report the acute kinematic and kinetic effects (Table I), followed by the acute performance effects and longitudinal effects (Table II).

Acute effects on sprint running kinematics

A total of eight investigations have assessed the acute effects of lower limb WR on movement kinematics when sprint running (see Table I). Four investigations loaded the whole leg by attaching WR to the thigh and shank (Bennett, Sayers, & Burkett, 2009; Macadam et al., 2017b; Simperingham & Cronin, 2014; Simperingham, Cronin, Pearson, & Ross, 2016), two investigations utilised a thigh loading scheme and a separate shank loading scheme (Feser, Macadam, Nagahara, & Cronin, 2018; Hurst, Johnston, Cronin, & Bezodis,

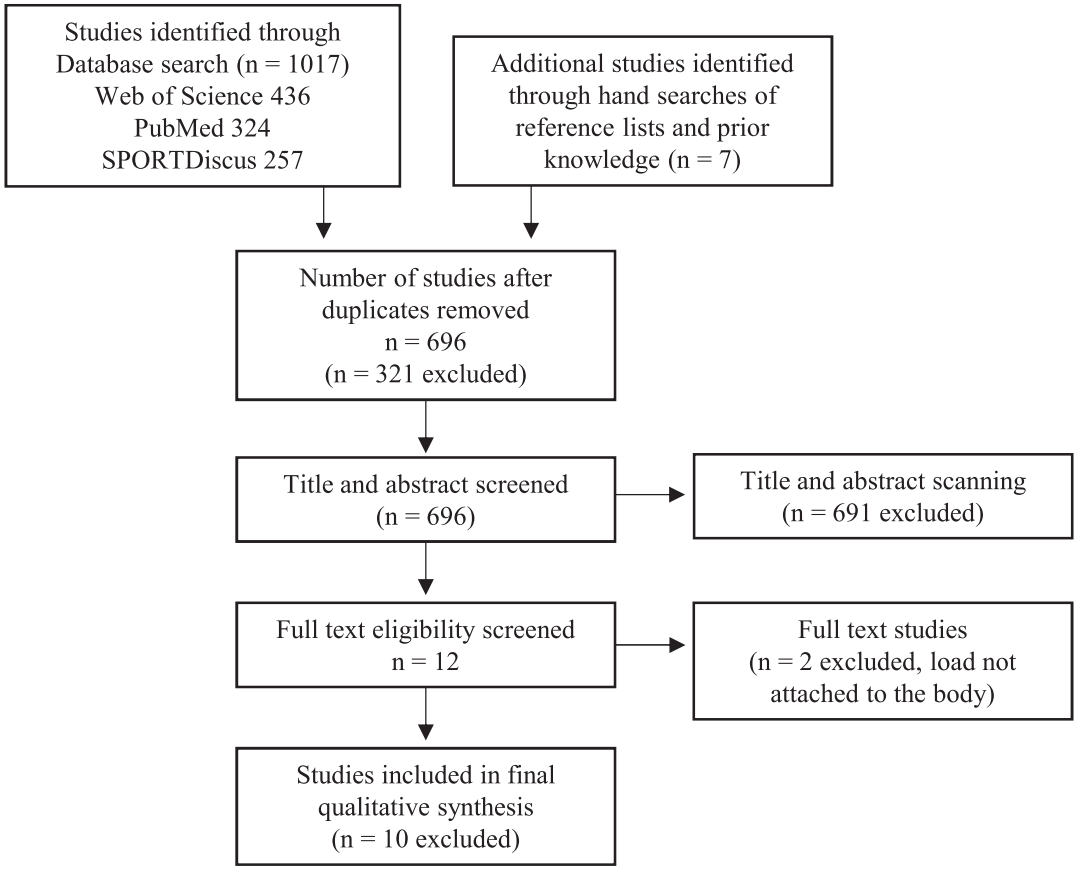

Figure 1. Flow of information through the different phases of the review.

2018), while one other study utilised a load placement at the ankle (Ropret et al., 1998). The following sections will summarise the kinematic effects of lower limb WR on sprint times and velocity, the start, acceleration, and maximal velocity phases of sprint running, and compare differences in WR location and magnitude.

Sprint times and velocity. Five of the studies included in this review reported sprint times. The WR used in these studies consisted of 10% of individual segment mass (Bennett et al., 2009), 3% BM (Macadam et al., 2017b; Simperingham et al., 2016), and 5% BM (Simperingham & Cronin, 2014; Simperingham et al., 2016) placed on the thigh and shank, 2% BM placed on the thigh (Feser et al., 2018), and 2% BM placed on the shank (Feser et al., 2018). The only statistically significant difference in total sprint times from the unloaded condition was found when loads of 5% BM were utilised. Specifically, Simperingham and Cronin (2014) reported the average 25 m non-motorised treadmill sprint time was increased 3.3% (ES = 0.43) with 5% BM WR. Simperingham et al. (2016) reported the average 20 m over ground sprint time to be increased 2.0% (ES = 0.37) with 5% BM WR. The latter was also a significant increase from the 3% BM condition used in Simperingham et al. (2016). Use of 3% BM in the Simperingham et al. (2016) study did not result in a significant difference in total sprint time from the unloaded condition which was consistent with other studies that utilised WR of 3% BM (Macadam et al., 2017b) or less (Bennett et al., 2009; Feser et al., 2018). Two studies also estimated the theoretical maximal velocity, which was reported to be significantly changed from the unloaded conditions in Simperingham et al. (2016) by −3.6% (ES = 0.50) with 3% BM WR and −6.0% (ES = 0.83) with 5% BM WR and in Macadam et al. (2017b) by −6.5% (ES = 0.84) with 3% BM posterior load location WR and −5.4% (ES = 0.60) with 3% BM anterior load location WR.

Start phase kinematics. When assessing the effect of lower limb WR on the start phase (as defined as the first two steps), contact time has shown to be significantly increased by 3.4–5.0% (ES = 0.41–0.49) with 3% BM WR (Macadam et al., 2017b; Simperingham et al., 2016) and by 5.0% (ES = 0.48) with 5% BM WR (Simperingham et al., 2016) while any changes to flight time, step frequency, and step length were

Table I. Effects of limb loading on sprint performance and biomechanical variables reported as percent change from the baseline unloaded condition ($n = 8$)^

Author	Participants (sex and mean ± SD age, height, and mass) Training Status	Sprint Distance	WR Loading	Sprint Performance (distance, % change, ES)		Kinematic Results (variable, % change, ES)		Kinetic Results (variable, % change, ES)	
Ropret et al. (1998)	24 males, 20.1 ± 0.9 years, 179.6 ± 8.4 cm, 74.5 ± 9.8 kg PE students, 18 active in a variety of sports activities	30 m sprint, over ground	1.6% BM, 3.2% BM, 4.8% BM; Ankle			AP (4.8% BM): velocity MVP (4.8% BM): velocity	−7.8%** (NA) −12.8%** (NA)		
Bennett et al. (2009)	8 males, 26.0 ± 7.3 years, 177.3 ± 3.4 cm, 77.3 ± 3.9 kg National level competitive beach sprinters (athletes' 100 m PR was 88% of current world record)	40 m sprint, over ground	2.4% BM; Whole leg, at radius of gyration	0–10 m 10–20 m 20–30 m 30–40 m 40 m	−0.6% 4.2%** (1.6) 3.7% 7.4%** (2.2) 3.1%	MVP: Stride velocity CT FT SF	−4.7%** (1.8) 8.9% 0.8% −2.2%		
Simperingham and Cronin (2014)	8 males, 29.2 ± 3.8 years, 177.1 ± 7.5 cm, 81.8 ± 9.7 kg Athletic at least 2 years of experience playing sprint based team sports	6 s sprint, non-motorized treadmill	5% BM; Whole leg	2 m 5 m 10 m 15 m 20 m 25 m 10–20 m	−1.8% 0.0% 1.1% 2.1%* (0.26) 2.5%* (0.32) 3.3%* (0.43) 5.2%* (0.66)	AP: Peak velocity CT FT SPF SL MVP: Peak velocity CT FT SPF SL	−2.3%* (0.44) 4.3%* (0.66) 0.0% −3.7%* (0.63) 6.0% −5.3%* (0.66) 4.7%* (0.60) 4.7% −3.5%* (0.66) −1.7%	AP: F_v $F_v$mean F_h Pmax $F_v$rel $F_v$mean.rel MVP: F_v $F_v$mean F_h Pmax $F_v$rel $F_v$mean.rel	4.0%* (0.22) 4.1%* (0.23) 0.6% −0.8% −0.9% −0.8% 4.5%* (0.28) 4.1% 1.9% −3.1% −0.8% −1.5%
Simperingham et al. (2016)	15 male, 19.0 ± 0.5 years 181.2 ± 7.3 cm, 91.0 ± 17.4 kg Rugby union athletes	20 m sprint, over ground	3% BM; Whole leg	5 m 10 m 20 m *v0*	−1.5% −0.5% 0.6% −3.6%* (0.50)	SP: CT FT SPF SL	5.0%* (0.41) −19.4% −1.5% 0.8%	$F0_{rel}$ $Pmax_{rel}$ S_{Fvrel}	6.25% 1.2% 10.0%* (0.72)

(*Continued*)

Table I. Continued.

Author	Participants (sex and mean ± SD age, height, and mass) Training Status	Sprint Distance	WR Loading	Sprint Performance (distance, % change, ES)		Kinematic Results (variable, % change, ES)		Kinetic Results (variable, % change, ES)	
						AP:			
						CT	5.0%*		
						FT	(0.55)		
						SPF	−3.8%		
						SL	−2.0%*		
							(0.31)		
							0.0%		
			5% BM; Whole leg	5 m	0.7%	SP:		$F0_{rel}$	1.3%
				10 m	0.9%	CT	5.0%*	$P\text{max}_{\text{rel}}$	−4.2%
				20 m	2.0%*	FT	(0.48)	S_{Fvrel}	6.1%
				$v0$	(0.37)	SPF	−17.7%		
					−6.0%*	SL	−2.0%		
					(0.83)		0.0%		
						AP:			
						CT	6.0%*		
						FT	(0.71)		
						SPF	−3.8%		
						SL	−3.0%*		
							(0.52)		
							−0.6%		
Macadam et al. (2017b)	19 male, 19.7 ± 2.3 years, 181 ± 6.5 cm, 96.1 ± 16.5 kg Amateur to semi-pro rugby, 2+ years sprint training and currently engaged in periodized S&C programme	20 m sprint, over ground	3% BM; Whole leg, anterior	2 m	−1.2%	SP:		$F0$	5.4%
				5 m	0.0%	CT	3.4%*	$P\text{max}$	1.5%
				10 m	0.0%	FT	(0.41)	S_{Fv}	12.2%*
				20 m	0.8%	SPF	−7.3%	$F0_{rel}$	(0.46)
				10–20 m	2.2%*	SL	−1.3%	$P\text{max}_{\text{rel}}$	5.5%
				$v0$	(0.35)	VS	0.0%	S_{Fvrel}	0.0%
					−5.4%*		−6.4%*		9.9%*
					(0.60)		(0.38)		(0.62)
						AP:			
						CT	3.0%*		
						FT	(0.39)		
						SPF	1.3%		
						SL	−3.4%*		
						VS	(0.42)		
							0.0%		
							−5.8%		
			3% BM; Whole leg, posterior	2 m	−1.2%	SP:		$F0$	4.7%
				5 m	0.0%	CT	4.4%*	$P\text{max}$	0.6%
				10 m	0.5%	FT	(0.49)	S_{Fv}	11.7%*
				20 m	1.4%	SPF	−7.3%	$F0_{rel}$	(0.51)
				10–20 m	2.9%*	SL	−1.3%	$P\text{max}_{\text{rel}}$	5.1%
				$v0$	(0.50)	VS	0.8%	S_{Fvrel}	−1.3%
					−6.5%*		−8.3%*		9.9%*
					(0.84)		(0.50)		(0.62)

						AP:	
						CT	3.0%*
						FT	(0.41)
						SPF	2.5%
						SL	−3.6%*
						VS	(0.44)
							0.0%
							−6.2%*
							(0.38)
Feser et al. (2018)	11 male, 21.2 ± 2.6 years, 175.3 ± 5.5 cm, 68.7 ± 4.3 kg Track and Field amateur athletes with 9.7 ± 2.9 years of sprint-based training	50 m sprint, over ground	2% BM; Shank, mid	10 m	−0.5%	Peak velocity	−1.8%*
				50 m	−0.2%		(0.37)
						AP:	
						CT	2.0%
						FT	2.0%
						SPF	−2.1%*
						SL	(0.34)
						SW	0.0%
							0.0%
						MVP:	
						CT	2.9%*
						FT	(0.42)
						SPF	3.3%*
						SL	(0.42)
						SW	−2.5%*
							(0.52)
							−0.5%
							0.0%
			2% BM; Thigh, distal	10 m	0.5%	Peak velocity	−2.0%*
				50 m	0.3%		(0.40)
						AP:	
						CT	2.6%
						FT	0.0%
						SPF	−1.4%
						SL	0.0%
						SW	0.0%
						MVP:	
						CT	3.8%**
						FT	(0.57)
						SPF	0.0%
						SL	−1.4%*
						SW	(0.31)
							−0.5%
							11.1%

(*Continued*)

Table I. Continued.

Author	Participants (sex and mean ± SD age, height, and mass) Training Status	Sprint Distance	WR Loading	Sprint Performance (distance, % change, ES)	Kinematic Results (variable, % change, ES)	Kinetic Results (variable, % change, ES)
Hurst et al. (2018)	6 male and 2 female, 21 ± 1 years, 172 ± 9 cm, 70.4 ± 6.4 kg University level sprinters	40 m sprint, over ground	0.6% BM; Shank, anterior at +4.5% moment of inertia		MVP: Step velocity CT FT SPF SL	−1.3%†† (0.18) 1.7%†† (0.17) 2.7%†† (0.26) −2.2%††† (0.34) 1.0%†	
			1.7% BM; Thigh, anterior at +4.5% moment of inertia		MVP: Step velocity CT FT SPF SL	−1.8%†† (0.23) 2.6%††† (0.26) 4.6% −3.8%††† (0.56) 1.5%	
Zhang, Yu, and Liu (2018)	16 males, 21 ± 2 years, 176 ± 4 cm, 67.41 ± 5.72 kg Sub-elite sprinters	40 m sprint, over ground	0.6% BM; Shank, 39.3% of shank length away from knee joint centre		MVP Velocity Landing distance Landing height Take-off distance Take-off height Hip joint LA Knee joint LA Ankle joint LA Hip joint TA Knee joint TA Ankle joint TA	−2.2* (0.40) 5.3% 0.0% −3.7% 0.0% 1.9% −5.1%* (0.28) 5.5% 16.5% −1.8% −12.7%	

^SD = standard deviation; ES = effect size; BM = body mass; NA = not enough information provided to calculate effect size; CT = contact time; SF = stride frequency; SPF = step frequency; SL = step length; VS = vertical stiffness; SP = start phase; AP = acceleration phase; MVP = maximal velocity phase; F_v = vertical ground reaction force; F_h = horizontal ground reaction force; Pmax = maximal horizontal power; rel = relative to system mass; $v0$ = theoretical maximal velocity; $F0$ = theoretical maximal horizontal force; S_{Fv} = slope of the force-velocity curve; WR = wearable resistance; *$p < 0.05$; **$p < 0.01$; †Possible trivial difference; ††Possible clear difference; †††Likely clear difference; LA = angle at landing; TA = angle at takeoff.

non-significant. Macadam et al. (2017b) also reported vertical stiffness to be significantly reduced by −6.4% (ES = 0.38) and −8.3% (ES = 0.50) with anterior and posterior 3% BM WR, respectively.

Acceleration phase kinematics. Lower limb WR appears to have a slightly greater effect on the acceleration phase (defined as steps 3–8) (Macadam et al., 2017b; Simperingham et al., 2016) or steps 3–12 (Simperingham & Cronin, 2014) compared to the start phase of sprint running as researchers have reported a significant decrease in step frequency in addition to significant increases to contact times when the WR magnitude is equal to 3% BM or greater (Macadam et al., 2017b; Simperingham & Cronin, 2014; Simperingham et al., 2016). An exception to this was Feser et al. (2018) who found 2% BM WR placed on the shank to significantly decrease step frequency (−2.1%, ES = 0.34) with no corresponding significant increase to contact time. Contact time was reported to be increased by 3.0% (ES = 0.39–0.41) with 3% BM WR (Macadam et al., 2017b) and by 4.3% (ES = 0.66) and 6.0% (ES = 0.71) with 5% BM WR compared to the unloaded conditions (Simperingham & Cronin, 2014; Simperingham et al., 2016). Step frequency was reported to significantly decrease with both 3% and 5% BM WR by −2.0 to −3.6% (ES = 0.31–0.44) and −3.0 to −3.7% (ES = 0.52–0.63), respectively, as compared to the unloaded condition (Macadam et al., 2017b; Simperingham & Cronin, 2014; Simperingham et al., 2016). While step length changes remained non-significant through the acceleration phase, the significant step frequency changes resulted in significant split time differences with 3% and 5% BM WR. These split time increases appeared significantly different from the unloaded condition at the 15 m (Simperingham & Cronin, 2014), 20 m mark (Simperingham et al., 2016), and from 10 to 20 m (Macadam et al., 2017b).

Simperingham and Cronin (2014) also reported peak velocity during the acceleration phase which was found to be significantly reduced by −2.3% (ES = 0.44) with 5% BM WR. Vertical stiffness measures were reported to be significantly lower, by −6.2% (ES = 0.38), when the 3% BM WR was positioned on the posterior leg (thigh and shank) but the corresponding decrease in stiffness when the 3% BM WR was positioned on the anterior surface of the leg was non-significant (−5.8%) (Macadam et al., 2017b). It is important to note that no other variables were found to be significantly different from the unloaded condition due to the anterior versus posterior loading nor was there any statistical differences found between the anterior and posterior load positions (Macadam et al., 2017b).

Maximal velocity phase kinematics. Researchers from six different studies have reported kinematic results for the maximal velocity phase of sprint running. Velocity was found to be significantly decreased (or decreased by a clear possible difference) by −1.3% to −12.8% in all studies (Bennett et al., 2009; Feser et al., 2018; Hurst et al., 2018; Ropret et al., 1998; Simperingham & Cronin, 2014; Zhang et al., 2018). All but one research group found step frequency and contact times to be significantly reduced or reduced by a clear possible difference/likely clear difference by −1.4% to −3.5% and 1.7% to 4.7%, respectively, with WR placed on the shank (Feser et al., 2018; Hurst et al., 2018), thigh (Feser et al., 2018; Hurst et al., 2018), or across the whole leg (Simperingham & Cronin, 2014). Flight time was also found to be changed ($p < 0.05$) with shank WR by 3.3% (ES = 0.42) with 2% BM WR (Feser et al., 2018) and possible clear difference of 2.7% with 0.6% BM WR (Hurst et al., 2018).

Additionally, researchers from two studies measured changes in joint kinematics with lower limb WR. Bennett et al. (2009) reported no differences ($p < 0.05$) between the 2.4% BM whole leg WR condition and the unloaded condition for sagittal plane hip and knee joint angular displacement or velocity variables. Zhang et al. (2018) found no significant changes to take-off and landing hip, knee, and ankle angles, distance, or height. However, knee joint landing angle was significantly decreased by 5.1% (ES = 0.28) (more knee extension) with shank WR. Furthermore, Zhang et al. (2018) found that although the WR used in their study significantly reduced sprint running velocity, it did not significantly change the relationship of sprint running velocity with the measured kinematic variables at landing and take-off.

Acute effects on sprint running kinetics

Three studies have reported the kinetic effects of lower limb WR (Macadam et al., 2017b; Simperingham & Cronin, 2014; Simperingham et al., 2016). Each of these studies utilised a 3% or 5% BM whole limb (thigh and shank) WR scheme. Those that investigated the effect of 3% BM WR on sprint kinetics found significant changes to the relative force-velocity (*F-v*) profile of 10.0–11.0% reflecting a more force dominant *F*-v profile (ES = 0.62–0.72) (Macadam et al., 2017b; Simperingham et al., 2016). Furthermore, Simperingham et al. (2016) found a significant increase of 9.0% in F_0 when expressed relative to body mass. When comparing kinetic outputs during the start, acceleration, and maximal velocity phase, Simperingham and Cronin (2014) reported no differences ($p < 0.05$) for mean vertical ground reaction force values between

Table II. Acute performance and longitudinal effects of limb loading on sprint performance and biomechanical variables ($n = 4$)^

	WR Loading	Participants (sex and mean ± SD age, height, and mass)	Training Protocol	Sprint Performance (distance, % change)		Results (variable, % change)	
Acute performance effects							
Bennett et al. (2009)	Whole leg; 2.4% BM	8 males, 26.0 ± 7.3 years, 177.3 ± 3.4 cm, 77.3 ± 3.9 kg National level competitive beach sprinters	7 repetitions of 40 m sprints; 2 unloaded (pretest), 2 loaded, 3 unloaded (posttest).	0–10 m 10–20 m 20–30 m 30–40 m 40 m	0.0% 2.6% −1.8% 1.8% 0.3%	MVP: stride velocity CT FT SF	 −0.8% 2.2% −0.8% 0.4%
Simperingham and Cronin (2014)	Whole leg; 5% BM	8 males, 29.2 ± 3.8 years, 177.1 ± 7.5 cm, 81.8 ± 9.7 kg Athletic with at least 2 years of experience playing sprint based team sports	4 sets of 2 max effort 6 s sprints, non-motorized treadmill; Sets 1 and 4 unloaded, sets 2 and 3 torso or leg loading randomised			MVP F_v	 1.3%* (NA)
Simperingham et al. (2015)	Whole leg; 1%, 3%, 5% BM	1 male, 29.2 years, 180.8 cm, 87.2 kg Rugby union athletes	3 × 40 m loaded accelerations with 1%, 3%, 5% BM	10 m 30 m 40 m 30–40 m	−3.3%^ −0.7% 0.0% 1.8%	SP: CT FT SPF SL VS AP: CT FT SPF SL VS MVP: CT FT SPF SL VS	 2.9%^ −1.8% −1.4% 1.8% −5.0% 2.1%^ 1.2% −1.8%^ 1.3% −6.1% 1.9% 3.2%^ −2.5%^ 0.5% −1.5%
Longitudinal effects							
Pajic et al. (2011)	Ankle; 5% BM	6 individuals, gender not specified 20.4 ± 1.7 years, 178.4 ± 8.12 cm, 71.4 ± 8.5 kg Untrained	6 week duration; 5 repetitions of 50 m sprints; increased to 2 sets for weeks 3 and 4, increased to 3 sets for weeks 5 and 6			MVP: velocity SF SL	 −0.5% −5.6%** (0.54) 5.3%** (0.43)

^SD = standard deviation; ES = effect size; BM = body mass; NA = not enough information provided to calculate effect size; CT = contact time; SF = stride frequency; SPF = step frequency; SL = step length; VS = vertical stiffness; SP = start phase; AP = acceleration phase; MVP = maximal velocity phase; WR = wearable resistance; *$p < 0.05$; **$p < 0.01$; ^More than 2 standard deviations from baseline mean.

the unloaded and WR conditions during the start and maximal velocity phase while there was a significant increase during the acceleration phase.

Differences between WR locations and magnitudes

Regarding lower limb WR sprint running kinematics, two studies have assessed the effects of differences between WR locations and one study assessed the effects of differences between WR magnitudes. Concerning differences in placement, Macadam et al. (2017b) found no statistically significant differences when 3% BM WR was placed on the anterior versus the posterior of the leg for any of the variables measured. Feser et al. (2018) reported a significant difference for flight time when 2% BM WR was placed on the shank versus the thigh. Specifically, the shank WR produced a 3.3% (ES = 0.47) increase in flight time over the thigh WR condition.

When the magnitude of WR was increased from 3% to 5% BM, a significant difference in 20 m sprint times and maximal horizontal power output was found. The 5% BM WR condition produced slower (−1.4%) sprint times and lower (−5.8%) relative maximal horizontal power output values than the 3% BM condition (Simperingham et al., 2016). Neither of these WR conditions produced a significant difference from the unloaded condition; the 3% BM WR produced a slight increase (1.2%) and the 5% BM WR produced a decrease (−4.2%) in relative maximal horizontal power output. Furthermore, a significant change (10.0%, ES = 0.72) in the slope of the *F*-v curve was found with 3% BM WR but not with 5% BM WR (−6.1%). Therefore, both WR loads resulted in a more force-oriented F-v curve but only 3% BM WR produced a significant change from the unloaded condition.

Acute performance and longitudinal effects

The effects following lower limb WR use on sprint running performance have only been minimally investigated. Three studies measured the changes that occur to sprint running performance immediately following a single session lower limb WR intervention (referred to as acute performance effects) (Bennett et al., 2009; Simperingham & Cronin, 2014; Simperingham, Cronin, Pearson, & Ross, 2015). One study measured the changes that occur to sprint running performance following a six week long lower limb WR intervention (Pajic et al., 2011).

Researchers that measured the changes to sprint running performance following the use of lower limb WR reported no significant change to total sprint times measured up to 40 m (Bennett et al., 2009; Simperingham et al., 2015). Simperingham et al. (2015) found during a single-subject study design that start and acceleration phase kinematics to be substantially changed (greater than two standard deviations from baseline) following a series of 40 m sprints completed with 1%, 3%, and then 5% BM whole leg WR. More specifically, contact time was substantially increased by 2.9% (start phase) and 2.1% (acceleration phase) with a substantial decrease in 10 m sprint time by −3.3%. During the maximal velocity phase, Simperingham et al. (2015) found substantial changes to both flight time (3.2%) and step frequency (−2.5%) with only a slight increase to the 30–40 m split time (1.8%). Bennett et al. (2009) reported no significant changes to 40 m sprint times following a 2.4% BM whole leg WR protocol while Simperingham and Cronin (2014) reported that a 5% BM WR protocol did not result in significant changes to running performance aside from a 1.3% increase in vertical ground reaction force production.

A six week training study utilising 5% BM ankle WR was found to elicit a significant increase in stride length (5.3%, ES = 0.43) and decrease in stride frequency (−5.6%, ES = 0.54) as measured between 25 and 50 m with no significant changes to maximal running speed (Pajic et al., 2011). Furthermore, this training was found to have significantly increased all angles measured (knee angle at take-off, knee angle at contact, knee flexion during backswing, and elbow flexion during backswing) suggesting a more upright body posture. The control group did not experience any statistically significant kinematic changes following the training.

Discussion

A specific concern with resisted sprint training is that the added load can unfavourably change sprint kinematic variables such as stride or step length, ground contact times, and stride or step frequency during the training. The micro-loading induced by the WR studies completed to-date (0.6%–5% BM) appeared to consistently affect some of the commonly measured kinematic variables (in particular step frequency and contact time) when the load was partitioned across the whole leg, thigh, or shank. When the load was solely located on the end of the lower limb (at the ankle), thus inducing a greater increase in limb inertia, an estimated 4.8% BM load resulted in a significant decrease in sprint velocity though stride length was unaffected. Practitioners may be interested in utilising higher loads, to the extent deemed comfortable by the athlete, to induce a greater loading stimulus, especially if looking to overload the start phase specifically. While step kinematic changes appeared to be cumulative from the start to the maximal velocity phase, participants appear to be able to maintain stride length which may further suggest joint kinematics are minimally affected when sprint running with an added 5% BM load on the lower limbs. This may alleviate practitioner concerns that training with lower limb WR negatively affects movement technique.

Many different lower limb WR configurations and magnitudes can be used during sprint running. Researchers thus far have investigated the effects of whole limb, thigh, shank, or ankle WR on sprint performance. When comparing the different WR location options, it appears the effects on sprint running performance were consistent. Specifically, all reports found that running velocity was affected as a result of the concomitant change in stride rate but not stride length (Bennett et al., 2009; Feser

et al., 2018; Hurst et al., 2018; Macadam et al., 2017b; Ropret et al., 1998; Simperingham & Cronin, 2014; Simperingham et al., 2016), with the exception of a possible trivial difference found by Hurst et al. (2018) for step length with shank WR. This, along with the lack of significant changes to joint kinematics found in Bennett et al. (2009) and minimal changes to joint kinematics found in Zhang et al. (2018) (all effect sizes under 0.28) reiterates the possibility that lower limb WR up to 5% BM could be used without significant coinciding changes to joint movements. Though further research is needed into this area.

Practitioners interested in WR should also understand the kinetic responses to limb loading to best determine how the loading stimulus can be utilised as a sprint running training tool. WR of 3% BM was found to elicit moderate changes (10.0–11.0%, ES = 0.62–0.72) to the relative *F*-*v* profile, reflecting a more force dominant profile (Macadam et al., 2017b; Simperingham et al., 2016). Furthermore, these researchers noted that F_0 was increased by ~5.0%–9.0% (Macadam et al., 2017b; Simperingham et al., 2016). Although F_0 has not been shown to be significantly correlated to sprint performance (i.e. max speed, mean 100 m speed (Morin et al., 2012)) it has been reported that sprinting expertise through the acceleration phase is characterised by an ability to produce higher amounts of horizontal net force at each step (Morin et al., 2012; Morin, Edouard, & Samozino, 2011; Rabita et al., 2015). Together these findings indicate that a 3% BM lower limb WR may provide a stimulus to increase horizontal force output during 20 m sprints and, therefore, it may be concluded that lower limb WR has the potential to elicit improved sprinting performance over time through greater horizontal force production.

It is important to note that sprinting expertise through the acceleration phase is marked by an ability of the athletes to apply more horizontally oriented (not greater) forces (Morin et al., 2012; Morin et al., 2011; Rabita et al., 2015). It is unclear if the participants included in the studies reviewed here responded to the loading by producing greater horizontal forces or merely reoriented the resultant force vector more horizontally as ground reaction force values were not directly measured in these studies, nor are enough details presented to make a conclusion. However, there remains the possibility that practitioners may be able to utilise lower limb WR to encourage more horizontally directed force production during the acceleration phase (via increased total force production or a reorientation of the force vector). It can also be concluded that the 3% BM WR successfully provided an overload stimulus for sprint running as theoretical maximal velocity values were significantly reduced with WR (Macadam et al., 2017b; Simperingham et al., 2016). This is an important factor considering one of the mechanical determinants of high levels of acceleration and 100 m sprint performance is a velocity oriented *F*-v profile (Morin et al., 2012). Considering the relationship of maximal horizontal force production and maximal running velocity to an individual's maximal mechanical power capabilities, a training programme intended to improve sprint acceleration performance should focus on improving both components of maximal horizontal power production (Morin & Samozino, 2016).

Previously researchers have suggested the value of high vertical force production to achieving fast speeds during maximal velocity running (Morin et al., 2012; Morin et al., 2011; Weyand, Sternlight, Bellizzi, & Wright, 2000). From our results it seems that a 5% BM lower limb WR may not overload the system enough to provoke greater vertical force production values during maximal velocity running as Simperingham and Cronin (2014) reported no significant differences for mean vertical ground reaction force values between the unloaded and WR conditions during the maximal velocity phase. However, this is in contrast to the significant increase in mean vertical ground reaction force values that were produced during the acceleration phase, further indicating that the effect of a given WR load varies for each phase of sprint running.

Minimal information has been reported on the acute performance effects of lower limb WR use for sprint running. From the literature reviewed it would seem that the removal of WR can produce immediate changes to sprint running kinematics with negligible effects to total sprinting time. How long these effects last is unknown. Due to limited research into acute performance effects with lower limb WR use, further research is required to assess how protocols affect the different phases of sprint running to find an optimum loading scheme and distance protocol.

Only one longitudinal study has investigated the effects of lower limb WR training on sprint performance. It was observed that 5% BM ankle WR elicited a significant increase in stride length (5.3%, ES = 0.43) and decrease in stride frequency (−5.6%, ES = 0.54) with non-significant changes (ES = 0.12) to maximal running speed (Pajic et al., 2011). While increases in running speed over time have been shown to coincide with an increase in step length and is believed to be important for development of maximal speed sprinting (Nagahara & Zushi, 2017) the accompanying decrease in stride frequency seen in Pajic et al. (2011) is counterproductive to the

development of maximal speed sprinting. Ultimately, it is difficult to apply these study findings as untrained participants were used and there is no information regarding the start and acceleration phases of sprinting. Additionally, with only one post-test time period completed any effects of off-loading the WR are unknown.

Given the paucity of research regarding the utilisation of lower limb WR for sprint running, further research is needed to better understand the effects of this type of training on the mechanical determinants of sprint performance and how WR can be best integrated into a sprint training regime. The studies that have been published to-date on the effects of lower limb WR on sprint running kinematics have only provided minimal information on changes to body posture and lower limb angular kinematics with no information presented during the start or acceleration phases of the sprint. Recently researchers have shown that lower limb angular kinematic changes are associated with improved sprint performance (Bezodis, North, & Razavet, 2017) and it has long been accepted that technical ability is of high importance to sprint running performance over simply being able to dynamically produce large forces (Morin et al., 2011). Therefore, it is of value to further investigate if lower limb WR can be used to induce positive changes to joint angular kinematics and total body posture in relation to sprint performance through the start, acceleration, and maximal velocity phases.

Additional evidence on the acute effects of lower limb WR on sprint running kinetics would be valuable to further understand how the WR load may be used to target certain aspects of sprint performance for training. Limited information has been published regarding force production values. The addition of information related to the index of force application and ratio of horizontal force to total force (as seen in Rabita et al., 2015) would be necessary to determine if athletes simply reorient force vectors with WR or if the WR stimulates an increase in total force production. Finally, more studies are needed to understand the utility of lower limb WR as a longitudinal training intervention for sprint performance. Once the effects of this type of training on sprint performance is determined, further training interventions could consider targeting particular phases of sprint running (i.e. start, acceleration, maximal velocity) as it is possible that different load magnitudes and/or orientation may need to be used to provide a desired overload within each phase.

Lower limb WR is a form of resisted sprint training that allows for the athlete to perform the sprint specific movement patterns in an overloaded manner with no changes to step or stride length. WR may be used to target certain aspects of sprint running. Practitioners may choose to utilise lower limb WR for an athlete needing to selectively overload stride frequency or encourage more horizontally directed force production during short distance sprint running or for the acceleration phase of longer distance sprint running. To target increased force production through the maximal velocity phase of sprinting, it is likely loads greater than 5% BM are needed to overload the system in the vertical direction. Lastly, considering the increased rotational inertia from WR, practitioners must take care to progressively overloading the athlete, especially during the beginning of the training season, to reduce possible risk of injury.

Disclosure statement

No potential conflict of interest was reported by the authors.

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